

Terminal-Lag Geometric Localization for the Complete Upper Counterloop Band

Bin Wang

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Abstract

We resolve how the complete deterministic upper-counterloop budget is distributed relative to its terminal order. Let $y_k = (\beta_k R)^2$, $x = (\beta R)^2 > 1$, and $y_k = x \exp[-\log C_M/k + o(k^{-1})]$. For

$$B_k(L) = \sum_{j=1}^L \frac{y_k^{k+j}}{k+j}, \quad C_k = B_k(k-1), \quad P_k(q) = B_k(k-1-q),$$

where $0 \leq q \leq k-2$, we prove uniformly in q that

$$\frac{P_k(q)}{C_k} = x^{-q} C_M^{q/k} \frac{2k-1}{2k-1-q} \frac{1-x^{-(k-1-q)}}{1-x^{-(k-1)}} (1+o(1)).$$

The corresponding exact terminal-lag probability has weights proportional to $y_k^{-r}/(2k-1-r)$, $0 \leq r \leq k-2$. Extended by zero to the nonnegative integers, it converges in ℓ^1 to $(1-x^{-1})x^{-r}$, and its mean and variance converge to $1/(x-1)$ and $x/(x-1)^2$. Thus a fixed terminal window of width q retains mass $1-x^{-q}$, and vanishing relative truncation error occurs exactly when $q \rightarrow \infty$. We also separate the sublinear, linear, and fixed residual-depth regimes. Actual-head inheritance is stated only conditionally on the original unnormalized same-clock leaf $D_{4k}(R) \rightarrow 0$. No root or rank identification, determinant closure, Gate A–E promotion, Hilbert–Polya construction, zero identification, or proof of RH is claimed.

1 Source-locked terminal ledger

Fix the physical constants

$$r_H = \frac{17}{20}, \quad R = \frac{7}{5}, \quad \frac{28}{17} < \lambda < \frac{17}{10}. \quad (1)$$

Let $k = k_\sigma \rightarrow \infty$ be the first-alias clock used in RH-342. The boundary multiplier and Hardy normalization are

$$|M_k| = C_M \lambda^k \{1 + o(1)\}, \quad C_M > 0, \quad (2)$$

$$\beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}. \quad (3)$$

Consequently

$$x = (\beta R)^2 = \frac{(R/r_H)^2}{\lambda} > 1, \quad y_k = (\beta_k R)^2 = x \exp \left[-\frac{\log C_M}{k} + o(k^{-1}) \right]. \quad (4)$$

These facts are the source locks from RH-17 and RH-342 [3, 7].

The deterministic graded counterloop is

$$\mathcal{Y}_k = \{\beta_k e^{ij\pi/k}, \beta_k e^{-ij\pi/k} : 1 \leq j \leq k-1\}, \quad (5)$$

with exact power ledger

$$s_{k,n} = \sum_{\nu \in \mathcal{Y}_k} \nu^n = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n). \quad (6)$$

The even strict upper band $2k < n < 4k$ therefore has budget

$$B_k(L) := \sum_{j=1}^L \frac{|s_{k,2k+2j}| R^{2k+2j}}{2k+2j} = \sum_{j=1}^L \frac{y_k^{k+j}}{k+j}, \quad 1 \leq L \leq k-1. \quad (7)$$

This is the deterministic ledger isolated in RH-355–RH-357 [4–6].

Define the complete strict upper budget and its lower residual after removing the top q terminal coordinates by

$$C_k := B_k(k-1), \quad P_k(q) := B_k(k-1-q), \quad 0 \leq q \leq k-2. \quad (8)$$

The symbol q in this paper is only an integer terminal lag. It is not the open direct/full-trace quantity denoted by the same letter in other project ledgers.

Proposition 1.1 (Exact terminal probability identity). *For $0 \leq r \leq k-2$, put*

$$\pi_k(r) := \frac{y_k^{2k-1-r}/(2k-1-r)}{C_k}, \quad (9)$$

and set $\pi_k(r) = 0$ for $r \geq k-1$. Then π_k is a probability on $\mathbb{Z}_{\geq 0}$ and, for every $0 \leq q \leq k-2$,

$$\boxed{\frac{P_k(q)}{C_k} = \sum_{r=q}^{k-2} \pi_k(r)}. \quad (10)$$

Proof. In (7), substitute $r = k-1-j$. The full range $1 \leq j \leq k-1$ becomes $0 \leq r \leq k-2$, and

$$C_k = \sum_{r=0}^{k-2} \frac{y_k^{2k-1-r}}{2k-1-r}.$$

The partial range $1 \leq j \leq k-1-q$ becomes $q \leq r \leq k-2$. Normalization gives both assertions. $\square$

2 Uniform finite-tail profile

We first record the endpoint estimate in the form needed below.

Lemma 2.1 (Uniform endpoint localization). *Uniformly for $1 \leq L \leq k-1$,*

$$B_k(L) = \frac{y_k^{k+L+1}(1-y_k^{-L})}{(k+L)(y_k-1)} \left(1 + O(k^{-1})\right). \quad (11)$$

Proof. Reverse the sum in (7):

$$B_k(L) = \frac{y_k^{k+L}}{k+L} \sum_{r=0}^{L-1} y_k^{-r} \frac{k+L}{k+L-r}.$$

Choose $1 < \xi < x$ with $y_k \geq \xi$ eventually. Since $k+L-r \geq k+1$,

$$0 \leq \sum_{r=0}^{L-1} y_k^{-r} \left(\frac{k+L}{k+L-r} - 1 \right) \leq \frac{1}{k+1} \sum_{r \geq 0} r \xi^{-r} = O(k^{-1}).$$

The unweighted geometric sum is at least one and equals $y_k(1-y_k^{-L})/(y_k-1)$. This proves the relative estimate uniformly in L . $\square$

Theorem 2.2 (Uniform terminal-lag theorem). *Assume (4). Uniformly over all integers $0 \leq q \leq k - 2$,*

$$\frac{P_k(q)}{C_k} = y_k^{-q} \frac{2k-1}{2k-1-q} \frac{1-y_k^{-(k-1-q)}}{1-y_k^{-(k-1)}} \left(1 + O(k^{-1})\right), \quad (12)$$

$$= x^{-q} C_M^{q/k} \frac{2k-1}{2k-1-q} \frac{1-x^{-(k-1-q)}}{1-x^{-(k-1)}} (1 + o(1)). \quad (13)$$

Both errors are relative and uniform in q . In particular, the factor x^{-q} alone is not a uniform asymptotic over the full range.

Proof. Apply lemma 2.1 first with $L = k - 1 - q$ and then with $L = k - 1$. Taking the ratio gives (12), because the quotient of two uniform factors $1 + O(k^{-1})$ is again $1 + O(k^{-1})$.

For the source-locked replacement, define

$$\rho_k := k \log(y_k/x) + \log C_M \longrightarrow 0.$$

Then, uniformly for $0 \leq q \leq k - 2$,

$$y_k^{-q} = x^{-q} C_M^{q/k} \exp\left(-\frac{q}{k} \rho_k\right) = x^{-q} C_M^{q/k} (1 + o(1)). \quad (14)$$

It remains to replace the finite-tail factors. For $m \geq 1$, let $F_m(t) = 1 - t^{-m}$. On $t \geq \xi$,

$$|F'_m(t)| = m t^{-m-1} \leq \sup_{j \geq 1} j \xi^{-j-1} < \infty, \quad F_m(x) \geq 1 - x^{-1} > 0.$$

The mean-value theorem therefore gives $F_m(y_k)/F_m(x) \rightarrow 1$ uniformly in $m \geq 1$. Apply this with $m = k - 1 - q$ and $m = k - 1$ and combine with (14). $\square$

3 Three lag regimes and the truncation criterion

Corollary 3.1 (Sublinear, linear, and residual-depth laws). *Let $q = q_k$ be admissible.*

1. *If $q = o(k)$ (hence $k - q \rightarrow \infty$), then*

$$\frac{P_k(q)}{C_k} = x^{-q} (1 + o(1)). \quad (15)$$

In particular, for fixed q , $P_k(q)/C_k \rightarrow x^{-q}$.

2. *If $q/k \rightarrow \theta \in [0, 1)$, then*

$$\frac{P_k(q)}{C_k} \sim \frac{2C_M^\theta}{2-\theta} x^{-q}. \quad (16)$$

3. *If $q = k - 1 - \ell$ with a fixed integer $\ell \geq 1$, then*

$$\frac{P_k(k - 1 - \ell)}{C_k} \sim 2C_M x^{-(k-1-\ell)} (1 - x^{-\ell}). \quad (17)$$

Proof. Each statement follows by taking limits in (13). In the first case, $C_M^{q/k} \rightarrow 1$, $(2k-1)/(2k-1-q) \rightarrow 1$, and the finite-tail ratio tends to one. In the second case these factors tend to C_M^θ , $2/(2-\theta)$, and one. In the third case $k-1-q = \ell$, while $q/k \rightarrow 1$ and $(2k-1)/(2k-1-q) = (2k-1)/(k+\ell) \rightarrow 2$. $\square$

The top terminal window of width q has normalized retained mass

$$T_k(q) := \sum_{r=0}^{q-1} \pi_k(r) = 1 - \frac{P_k(q)}{C_k}, \quad (18)$$

where $T_k(0) = 0$. Thus a fixed window has $T_k(q) \rightarrow 1 - x^{-q}$, not one.

Proposition 3.2 (Sharp qualitative window criterion). *For any admissible integer sequence $0 \leq q_k \leq k - 2$,*

$$\frac{P_k(q_k)}{C_k} \rightarrow 0 \iff q_k \rightarrow \infty. \quad (19)$$

Equivalently, the terminal retained mass tends to one exactly for diverging window width.

Proof. The explicit factors in (13) are uniformly bounded apart from x^{-q} : $C_M^{q/k}$ stays in a fixed compact positive interval, $(2k-1)/(2k-1-q) < 2$, and the finite-tail quotient is bounded because its denominator tends to one. Hence $P_k(q)/C_k \leq Kx^{-q}$ for all large k and all admissible q , proving sufficiency.

If q_k does not tend to infinity, it has a subsequence bounded by a fixed integer. A further subsequence is constant, say $q_k = q_0$. The fixed-lag case of (15) then gives the positive limit x^{-q_0} , so the tail cannot tend to zero. $\square$

4 Geometric localization in total variation

Theorem 4.1 (Terminal-lag distribution and moments). *Extend π_k by zero to all $r \in \mathbb{Z}_{\geq 0}$ and define*

$$\pi(r) := (1 - x^{-1})x^{-r}, \quad r \geq 0. \quad (20)$$

Then

$$\|\pi_k - \pi\|_{\ell^1(\mathbb{Z}_{\geq 0})} \rightarrow 0, \quad d_{\text{TV}}(\pi_k, \pi) \rightarrow 0. \quad (21)$$

Moreover,

$$\mathbb{E}_{\pi_k} r \rightarrow \frac{1}{x-1}, \quad \text{Var}_{\pi_k}(r) \rightarrow \frac{x}{(x-1)^2}. \quad (22)$$

Proof. Multiplying numerator and denominator in (9) by $(2k-1)y_k^{-(2k-1)}$ gives

$$\pi_k(r) = \frac{a_{k,r}}{A_k^*}, \quad a_{k,r} := \mathbf{1}_{0 \leq r \leq k-2} y_k^{-r} \frac{2k-1}{2k-1-r}, \quad A_k^* := \sum_{s \geq 0} a_{k,s}. \quad (23)$$

For each fixed r , $a_{k,r} \rightarrow x^{-r}$. Choose $1 < \xi < x$ with $y_k \geq \xi$ eventually. Since $0 \leq r \leq k-2$ implies $(2k-1)/(2k-1-r) < 2$,

$$0 \leq r^m a_{k,r} \leq 2r^m \xi^{-r}, \quad m = 0, 1, 2. \quad (24)$$

The right-hand sequences are summable. Dominated convergence on the counting measure therefore gives, for $m = 0, 1, 2$,

$$\sum_{r \geq 0} r^m a_{k,r} \rightarrow \sum_{r \geq 0} r^m x^{-r}. \quad (25)$$

For $m = 0$, $A_k^* \rightarrow (1 - x^{-1})^{-1}$; dominated convergence also yields $\sum_r |a_{k,r} - x^{-r}| \rightarrow 0$. Normalizing proves the ℓ^1 assertion, and $d_{\text{TV}} = \frac{1}{2} \|\cdot\|_1$.

The cases $m = 1, 2$ in (25), divided by A_k^* , give convergence of the first two moments. For a geometric law with ratio $t = x^{-1}$, the mean is $t/(1-t) = 1/(x-1)$ and the variance is $t/(1-t)^2 = x/(x-1)^2$, proving (22). $\square$

Remark 4.2 (What localization does and does not say). The theorem localizes deterministic weighted trace mass near the terminal coordinate. It does not turn π_k into an eigenvalue distribution, a root counting measure, or a probability law for noisy dynamics.

5 Conditional actual-head inheritance

Let $\mathcal{H}_\sigma$ be the actual modulus-complete Hardy head of RH-342, with moments $h_{\sigma,n}$, and put $d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}$. The original unnormalized same-clock transport leaf is

$$D_{4k}(R) := \sum_{2 \leq n < 4k} \frac{|d_{\sigma,k,n}| R^n}{n} \rightarrow 0. \quad (26)$$

It is an open hypothesis, not a conclusion of this paper. For $0 \leq r \leq k-2$, define

$$w_{k,r}^{\mathcal{H}} := \frac{|h_{\sigma,4k-2-2r}| R^{4k-2-2r}}{4k-2-2r}, \quad C_k^{\mathcal{H}} := \sum_{r=0}^{k-2} w_{k,r}^{\mathcal{H}}, \quad P_k^{\mathcal{H}}(q) := \sum_{r=q}^{k-2} w_{k,r}^{\mathcal{H}}. \quad (27)$$

When $C_k^{\mathcal{H}} > 0$, let $\pi_k^{\mathcal{H}}(r) = w_{k,r}^{\mathcal{H}}/C_k^{\mathcal{H}}$ on this range and set it to zero for $r \geq k-1$.

Theorem 5.1 (Conditional inheritance only). *Assume (26) on one physical clock. Then, eventually, $C_k^{\mathcal{H}} > 0$ and*

$$\frac{C_k^{\mathcal{H}}}{C_k} \rightarrow 1, \quad (28)$$

$$\sup_{0 \leq q \leq k-2} \left| \frac{P_k^{\mathcal{H}}(q)/C_k^{\mathcal{H}}}{P_k(q)/C_k} - 1 \right| \rightarrow 0, \quad (29)$$

$$\sup_{0 \leq r \leq k-2} \left| \frac{\pi_k^{\mathcal{H}}(r)}{\pi_k(r)} - 1 \right| \rightarrow 0, \quad (30)$$

$$\sum_{r \geq 0} r^m |\pi_k^{\mathcal{H}}(r) - \pi_k(r)| \rightarrow 0, \quad m = 0, 1, 2. \quad (31)$$

Consequently the actual even-weight terminal distribution inherits the total-variation and first-two-moment conclusions of theorem 4.1 only under (26).

Proof. The reverse triangle inequality and the fact that every displayed order is below $4k$ give

$$\sum_{r=0}^{k-2} \left| w_{k,r}^{\mathcal{H}} - \frac{y_k^{2k-1-r}}{2k-1-r} \right| \leq D_{4k}(R). \quad (32)$$

Therefore $|C_k^{\mathcal{H}} - C_k| \leq D_{4k}(R)$ and, uniformly in q , $|P_k^{\mathcal{H}}(q) - P_k(q)| \leq D_{4k}(R)$. Now $C_k \rightarrow \infty$, and the smallest partial budget is

$$P_k(k-2) = B_k(1) = \frac{y_k^{k+1}}{k+1} \rightarrow \infty.$$

Thus both full and partial budgets have uniform relative error tending to zero, which proves (28)–(29).

The same hypothesis gives a stronger coordinatewise statement. Every deterministic weight obeys

$$\frac{y_k^{2k-1-r}}{2k-1-r} \geq \frac{y_k^{k+1}}{2k-1}, \quad 0 \leq r \leq k-2,$$

and each summand difference in (32) is at most $D_{4k}(R)$. Hence

$$\sup_{0 \leq r \leq k-2} \left| \frac{w_{k,r}^{\mathcal{H}}}{y_k^{2k-1-r}/(2k-1-r)} - 1 \right| \leq \frac{(2k-1)D_{4k}(R)}{y_k^{k+1}} \rightarrow 0.$$

Together with (28), normalization proves (30). The domination argument in (24)–(25) shows that $\sum_r r^m \pi_k(r)$ is uniformly bounded for $m = 0, 1, 2$. Multiplying this bound by the supremum in (30) proves (31). The case $m = 0$ is the ℓ^1 conclusion and hence the total-variation conclusion; $m = 1, 2$ transfer the mean and variance. No root, rank, or determinant information is transferred by this argument. $\square$

6 Finite protocol, scope, and next route

The artifact uses the exact rational fixture $\lambda = 5/3$, $C_M = 1$, hence $x = 2352/1445$, to check the exact tail identity, the full uniform error envelope, and finite-support probability normalization. A separate high-precision synthetic multiplier law uses the repository diagnostic values of λ and C_M to check the source-locked linear and fixed-residual-depth profiles. It also reports ℓ^1 , mean, and variance errors at increasing k . These rows reproduce proved formulas; they are not interval certificates or observations of an actual noisy head.

The next read-only candidate is RH-359: invert the terminal-tail law at logarithmic window widths and retain the integer floor phase in polynomial accuracy thresholds. That route remains deterministic until a typed physical transfer theorem is available.

RH-358 does not prove (26); identify actual roots, ranks, or spectra; close a determinant, p , the unrelated open q , or E_{off} ; close the moving noisy envelope of RH-241 [1]; or activate the gluing criterion of RH-288 [2]. Gates A–E remain false/open. There is no Hilbert–Polya operator, Riemann-zero identification, von Mangoldt trace theorem, zeta-divisor equality, or proof of the Riemann hypothesis.

References

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